

DATA-DRIVEN CONSTITUTIVE MODELING OF THE PROGRESSIVE SOFTENING IN ELASTOMERIC GELS WITH INTERPENETRATING NETWORKS

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ABSTRACT

Solid though hydrogels are a class of bio-inspired elastomeric material with the ability of high stretch-ability, toughness, and water absorption. Though hydrogels exhibit nonlinear features such as stress softening, permanent set, cyclic energy dissipation and induced anisotropy in their mechanical responses. Besides, gels used in biomedical devices and soft robots are designed to sustain continuous cyclic loading during their service-life. While most studies assume that the majority of stress softening occurs in the first cycle. However, experimental studies suggest that the material exhibits damage accumulation and stiffness reduction significantly at higher number of cyclic loading. This progressive stress softening can lead to an unpredicted response of the material in dynamic loading as it is a function of material stiffness. However, there are only a few constitutive models developed to address this accumulative damage in the elastomeric gels. Thus, here, we developed a data-driven method to predict the behavior of hydrogels subjected to a high number of cycles. The developed model can be particularly used to estimate material response in each cycle and its steady-state response due to three-dimensional deformation. The strain energy of the material is the same in all direction in the undeformed state and damage evolve in each direction based on its maximum deformation. Moreover, the energy of the material in each direction estimated through two parallel artificial neural network. The proposed model shows good agreement with cyclic uni-axial tensile test data in different cycles of the loading and steady-state response of material.

Introduction

Hydrogels are bio-inspired elastomers with high water contents. Tough hydrogels with mechanical properties close to soft tissues are revolutionary bio-compatible materials with several new ranges of applications. Their unique properties in terms of high-water content, simple chemical modifications, bio-compatibility, softness, and flexibility make them excellent candidates for a wide range of medical and pharmaceutical applications (e.g. drug delivery, spinal cord surgery, wearable sensors, and cartilage) [1, 2]. However, most conventional hydrogels exhibit a fragile mechanical response, which limits their practical utilization to non-load bearing applications [3]. In recent decades, several strategies have been utilized to improve the mechanical properties of gels including their stretch-ability and toughness [4, 5]. One of the promising strategies to enhance the mechanical properties of the hydrogels is interpenetrating multiple polymer networks in a composite matrix, which is first proposed by Gong et al. [3] as a double network hydrogel (DN gels). DN gels are synthesized by interpenetrating a loosely cross-linked, soft, and stretchable neutral network (second) in a highly cross-linked, brittle, and tough polyelectrolyte gel (first network). This combination results in an exceptional stretch-ability as much as 13 times of the initial length and high stiffness and strength of gel in comparison with conventional single-network hydrogel [6, 7]. In recent years, numerous DN gels have been developed to further enhance their mechanical strength and toughness using different polymer network structures and energy absorption mechanisms.

DN gels show a nonlinear mechanical behavior due to load

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including hyper-elastic response and permanent damage. Thus, their exceptional mechanical performance attracted attentions in studying the nature of these nonlinear behaviors in mechanical response in addition to the kinetics of swelling of hydrogels. DN gels consist of two sets of interpenetrated polymer networks and trapped water inside of the material, which result into a nonlinear J- and S-shape type behavior in the primary loading under large deformations. In addition, the presence of water in the gel leads to an incompressible material. Furthermore, DN gels show different nonlinear features such as significant stress softening in their first cycle of loading, known as Mullins effect, permanent deformation, necking instability, and progressive softening due to the cyclic loading. In recent years, constitutive modeling of DN gel considering the idealized Mullins effect and time-dependent viscoelastic behavior have been a matter of interest [8, 9, 10, 11]. In these models, the behavior of the material in consecutive cycles of loading is considered the same. Thus, our understanding of the nature of progressive stress softening and accumulated damage due to a high number of cycles in the gels remain sparse and inconclusive.

The gels used in different devices may be subjected to dynamic and cyclic loading in their lifetime. In this context, the lifetime prediction of the material due to fracture and fatigue is well studied in recent years. In these studies, a significant accumulation of damage in the response of the material is reported during the cyclic loading. As a result of damage accumulation, three parallel phenomena happen in the behavior of the material in each cycle, (i) stiffness reduction, (ii) increase of permanent deformation, and (iii) increase of stress softening. The schematic cyclic behavior of the DN gel is depicted in Fig. 1. In addition, the dissipation of energy in the material exponentially decreases as the number cycles increases. Thus, we assume that energy dissipation after the first cycle is negligible and the unloading-reloading is considered the same. Also, in most of the studies, it is assumed that damage in the material is a function of its maximum stretch experienced. However, the cyclic behavior of elastomers shows that the material response after unloading follows the primary loading after a stretch higher than the maximum stretch. The distance between this stretch and the maximum stretch experienced in the loading significantly increases in a higher number of cycles.

In this study, we assume that material is isotropic in its virgin states and damage evolves in each direction based on micro-deformation in that direction. This non-uniform damage evolution in the material results in a deformation/cyclic-induced anisotropy in the material. To this end, we utilized two parallel Artificial Neural Network (ANN) to take into account effect of both damages. The first ANN considers for damage in the first cycle and the second one is an decreasing damage ANN, which is function of number of cyclic load in the material. Combination of these two parallel sub-network leads to a constitutive model, which can predict the behavior of the material in a cyclic loading. The proposed model can simulate different nonlinear feature in gels

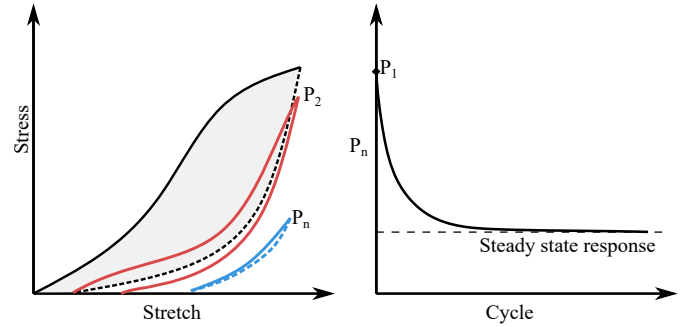


FIGURE 1. Constitutive response of DN hydrogel specimen under a cyclic loading, which includes three typical inelastic features, i.e., stress-softening, increasing permanent stretch, and stiffness reduction

including, increasing permanent set and decreasing stiffness of the material in both function of stretch and number of cycle.

This paper is organized as follows; in section 2, Artificial Neural Network is described briefly. The developed constitutive model based on ANN is presented in the next. In section 4, the model is tested against the experimental data. Finally, concluding remarks are presented in the section 5.

Artificial Neural Network

A set of artificial neuron inspired by the biological neural networks is utilized to mimic learning process in the brain. The collection of connected units and nodes form an artificial neural network (ANN). Each nodes in the ANN receives a signal its incoming connections and send its output based on its functionality as

$$y_i = \Phi(z_i), \quad z_i = \sum_j w_{ij}x_j, \quad (1)$$

where sets of x_j are neuron inputs, w_{ij} are connection weights, and Φ is its activation function. The output of the neuron is y_i . Activation Functions are a mathematical gate between input feeding of a neuron and its output going to the next layer. Activation function can be any different function such as *sigmoid*, *polynomial*, *tanh*, *tan*, *sin*, *cos*, *exp*, *log*, *RELU*. Activation function can be selected based on the application and nature of input and output data. Note that, each ANN contains different layers of input, hidden and output layer (see Fig. 2). Number of inputs, outputs should be defined based on the application and designed structure of the network.

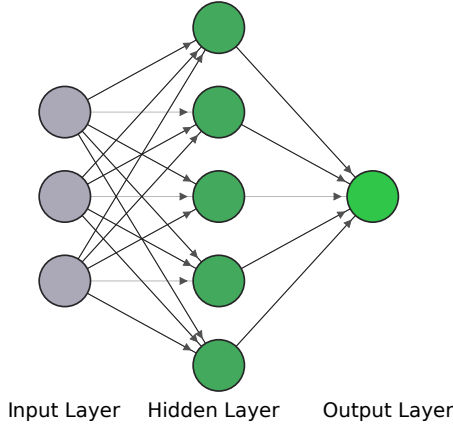


FIGURE 2. Schematic of an ANN including an input layer, hidden layers and an output layer

CONSTITUTIVE MODEL

Strain energy function of a nearly incompressible material $\Psi_N(\mathbf{C})$ can be written as (see e.g. [12])

$$\Psi_N(\mathbf{C}) = \Psi_M(\bar{\mathbf{C}}) + U(J), \quad (2)$$

where $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ and $\bar{\mathbf{C}} = J^{-\frac{2}{3}} \mathbf{C}$ are the right Cauchy-Green tensor and its distortional part respectively and $J = \sqrt{\det \mathbf{C}}$. Here, $\mathbf{F}$ is the deformation gradient, which can be decomposed into a elastic $\mathbf{F}_e$ and a swelling $\mathbf{F}_v = v_p^{-\frac{1}{3}} \mathbf{I}$ deformation tensors as $\mathbf{F} = \mathbf{F}_m \mathbf{F}_v$. The swelling deformation tensor $\mathbf{F}_v$ consider an uniform volume change in the sample during the water uptake or release as v_p . Here, we assume that material is a isotropic in a virgin state. Thus, the behavior of the material can be represented by considering same strain energy function for the material in all direction of the sample. This concept in the statistical mechanics of elastomers known as micro-sphere scale transition, which can be written as

$$\Psi_M = \frac{1}{A_s} \int_A \Psi_M^d d\mathbf{u} \cong \sum_{j=1}^k \Psi_M^{d_j} w_j, \quad (3)$$

where $A = 4\pi$ is the surface area of the unit sphere and d is the unit vector in normal direction to the surface area. $\Psi_M^{d_j}$ denotes the strain energy of the matrix and w_j is weight factor in direction d_j . Here, the numerical integration scheme with 21 integration points over half sphere is used, which is developed by Bazant and Oh [13].

The experimental studies suggest that strain energy function of gels evolve as function of maximum stretch and the number of cycles experienced [14]. Moreover, it is shown that damage in the material in DN gels is an anisotropic process [15]. Thus, here,

we assumed that the energy of matrix is the same in all directions in the virgin state. However, the damage non-uniformly evolves in the course of deformation and cyclic loading experienced in each direction. Thus, we assumed that two parallel Artificial Neural Network (ANN) are responsible for each type of damage in the material. The first ANN capture nonlinear behavior and damage of the material during first cycle including Mullins effect, deformation induced anisotropy and permanent set in the response of material. The second ANN is developed to capture progressive stress softening in material due to cyclic deformation. The schematic view of proposed model is depicted in Fig. 3. As it can be seen in this figure, the micro-mechanical concept of unit-sphere is combined with the a data-driven ANN to predict the behavior of the material during cyclic loading. The first ANN is developed based on the information of first cycle loading-unloading and can predict nonlinear features such as Mullins effect, permanent set and deformation induced anisotropy. The second ANN considers the progressive stress softening in each direction. Finally the total energy of the material can be calculated by multiplication of trained ANNs as

$$\Psi_M^{d_j} = D \Psi_0^{d_j} = ANN_2 \times ANN_1^{d_j}. \quad (4)$$

In the first ANN in each direction, three parameters are considered as inputs, representative of right Cauchy-Green tensor in direction d_j as λ^{d_j} , its maximum value over time $\lambda_{max}^{d_j}$ and representative of the Cauchy deformation tensor in direction d_j as β^{d_j} [16]. The second ANN is trained by maximum stretch of the cyclic loading $\lambda_c^{d_j}$, direction of the loading as a binary input and cycle number. In addition, in both ANNs an extra constant input is fed to the network to remove over-fitting and bias problems in ANN training. Output of first ANN is the response of the material before cyclic loading and output of the second ANN shows the amount of accumulation of damage in direction d_j .

The representative of right and left Cauchy-Green tensor in direction d_j , λ^{d_j} and β^{d_j} are defined as

$$\begin{aligned} \lambda^{d_i} &= \sqrt{\vec{d}_i \mathbf{C} \vec{d}_i}, \\ \beta^{d_i} &= \sqrt{\vec{d}_i \mathbf{c} \vec{d}_i}, \end{aligned} \quad (5)$$

where $\mathbf{c} = \mathbf{F}^{-T} \mathbf{F}^{-1}$ is the Cauchy deformation tensor. In addition, the maximal stretch $\lambda_{max}^{d_i}$ in the direction d_i can be written as

$$\lambda_{max}^{d_i} = \max_{\tau \in (-\infty, t]} \left[\lambda_p^{d_i}, \lambda^{d_i}(\tau) \right], \quad d_i \in \mathbb{V}^3 \wedge |d_i| = 1, \quad (6)$$

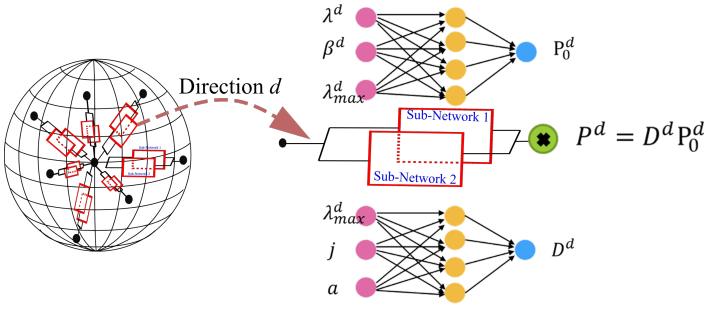


FIGURE 3. Schematic of proposed NN based constitutive model using micro-sphere concept and two damageable sub-networks

where λ_p is the maximum stretch experienced during polymerization and swelling. Finally, in view of continuum mechanic relation of strain energy and stress, the first Piola-Kirchhoff stress tensor $\mathbf{P}$ can be calculated as

$$\mathbf{P} = \frac{\partial \Psi_M}{\partial \mathbf{F}} - p \mathbf{F}^{-T}, \quad (7)$$

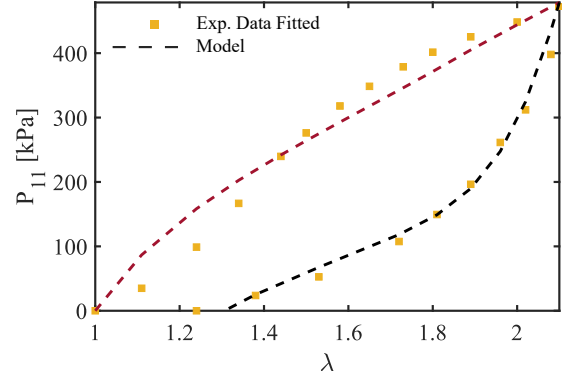
where p denotes hydro-static pressure to guarantee incompressibility. By substituting the total energy into Eq. 7, the first Piola-Kirchhoff stress tensor can be rewritten as

$$\frac{\partial \Psi_M}{\partial \mathbf{F}} = \sum_{i=1}^k w_i \frac{\partial \left(\frac{d_i}{D} \psi_0 \right)}{\partial \mathbf{F}} = \sum_{i=1}^k w_i \frac{d_i}{D} \frac{\partial \psi_0}{\partial \mathbf{F}}, \quad (8)$$

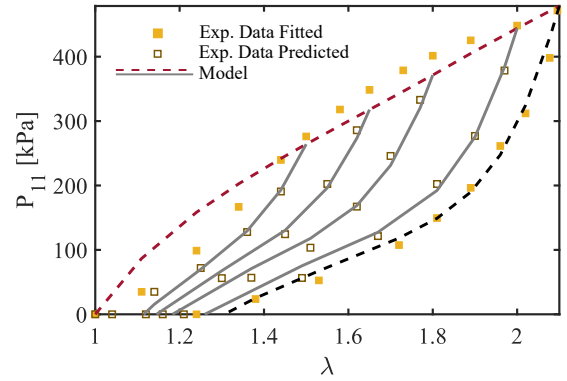
where

$$\begin{aligned} \frac{\partial \psi_0}{\partial \mathbf{F}} &= \frac{\partial \psi_0}{\partial \lambda} \frac{\partial \lambda}{\partial \mathbf{F}} + \frac{\partial \psi_0}{\partial \beta} \frac{\partial \beta}{\partial \mathbf{F}} \\ &= \frac{\partial \psi_0}{\partial \lambda} \frac{1}{\lambda} \mathbf{F} (d_i \otimes d_i) + \frac{\partial \psi_0}{\partial \beta} \frac{1}{\beta} \mathbf{F}^{-1} \mathbf{c} (d_i \otimes d_i). \end{aligned} \quad (9)$$

By means of Eq. 8, one can train each ANN by stress-stretch experimental data set. Here, we used a sequential training algorithm by assuming that there is not any accumulation of damage in the virgin material, $D|_{j=1} = 1$. Therefore, only first ANN is responsible for total response of the material. The first ANN is trained using the least-square error minimization approach for one cycle of loading and unloading data and similarly the second network is trained through the stress decay function at maximum stretch of cyclic loading. Note that activation function in each ANN can be selected based on the behavior of the material.



(a) First cycle of loading-unloading-data used for fitting



(b) First cycle of loading-unloading at different maximum stretch -model prediction

FIGURE 4. Comparison of the nominal stress-stretch curves of the model and the experiment for (a) the uni-axial tensile tests data used for fitting and (b) model prediction for unloading in different λ_{max}

EXPERIMENTAL VALIDATION

In order to validate the developed model to predict progressive stress softening, we used the experimental data from Zhang et al. [14]. This study focused on experimental investigation of DN gels uni-axial cyclic tensile test and damage accumulation in the material over thousands of cycles. Zhang et al. [14] used DN hydrogel, which were synthesized through sequential polymerization of the networks developed by Gong et. al. [3]. In the hydrogels used by Zhang et. al. [14], the first network was made from 1 M AMPS, the solution 4 mol% MBAA and 0.1 mol% OA relative to AMPS. The second network with 4 M AAM was synthesized afterward around the first swollen network (for details see [14]). In each test, a dumbbell-shaped specimen was elongated up to certain stretch level and then unloaded (one complete uni-axial tension cycle) and this procedure has been repeated up to twelve thousand times.

The first ANN is trained using first cycle of loading and unloading (Fig.4 (a)) in 21 directions. As can be seen in Fig.4 (b), model prediction has a good agreement with material behavior in

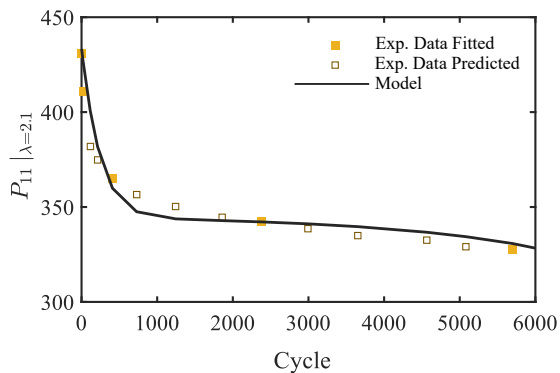


FIGURE 5. Comparison of the maximum nominal stress of the model and the experiment for as a function of number of loading cycles at $\lambda_c = 2.1$.

other cycles. The relative error of model is 4% for training set and 2.5% for the test set. In the next, step the train network along with experimental data of maximum stress during cyclic loading is used to train the second ANN. Here, only 5 data points of cyclic loading at $\lambda_c = 2.1$ are used to train the ANN. In Fig.5, model prediction of accumulative stress softening is plotted against experimental tests. As can be seen in Figs 4 and 5, the model prediction has good agreement with the experimental results.

CONCLUDING REMARKS

Here, a data-driven constitutive model is developed based on the concept of network decomposition and micro-sphere to describe the accumulation damage in quasi-static cyclic loading of DN gels. Two parallel ANN sub-networks are utilized in each direction. The first ANN sub-network considers the effect of damage in the first cycle as a function of maximum stretch experienced and the second ANN sub-network takes into account progressive stress softening due to cyclic loading. Each of these two sub-network evolves in each direction of micro-sphere, which results in deformation and vibration induced anisotropy and permanent set in the response of the material. Moreover, the model performance and goodness of its prediction is tested by training the model by only one set of loading and unloading data it to a set of experimental data. that shows its ability of describing both the Mullins effect and permanent set. The results show that the model has a good agreement with the experimental data. This performance makes the model a suitable option for commercial and other applications.

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